

Problem

Given a parallel RLC circuit with $R = 5 \text{ k}\Omega$, $L = 10 \text{ mH}$, and $C = 20 \text{ }\mu\text{F}$, if the circuit is magnitude-scaled by $K_m = 500$ and frequency-scaled by $K_f = 10^5$, find the resulting values of R , L , and C .

Step-by-step solution

Step 1 of 3

Consider a parallel RLC circuit having resistance R of $5 \text{ k}\Omega$, inductance L of 10 mH and capacitance C of $20 \text{ }\mu\text{F}$.

The circuit is magnitude scaled K_m by 500 and frequency scaled K_f by 10^5 .

Calculate the new values.

Calculate the new value of resistance R' .

Write the expression for the new value of resistance R' .

$$R' = K_m R$$

Substitute 500 for K_m , $5 \text{ k}\Omega$ for R in this expression.

$$\begin{aligned} R' &= (500)5 \times 10^3 \\ &= 2.5 \times 10^6 \end{aligned}$$

Therefore, the new value of resistance R' is $\boxed{2.5 \text{ M}\Omega}$.

Step 2 of 3

Calculate the new value of capacitance C' .

Write the expression for the new value of capacitance C' .

$$C' = \frac{1}{K_m K_f} C$$

Substitute 500 for K_m , $20 \text{ }\mu\text{F}$ for C , 10^5 for K_f in this expression.

$$\begin{aligned} C' &= \frac{1}{500(10^5)} 20 \times 10^{-6} \\ &= 0.4 \times 10^{-12} \end{aligned}$$

Therefore, the new value of capacitance C' is $\boxed{0.4 \text{ pF}}$.

Step 3 of 3

Calculate the new value of inductance L' .

Write the expression for the new value of inductance L' .

$$L' = \frac{K_m}{K_f} L$$

Substitute 500 for K_m , $20 \mu\text{F}$ for C , 10^5 for K_f in this expression.

$$\begin{aligned} L' &= \frac{500}{10^5} (10 \times 10^{-3}) \\ &= 50 \times 10^{-6} \end{aligned}$$

Therefore, the new value of inductance L' is $\boxed{50 \mu\text{H}}$.